# Transport anisotropy in spontaneously ordered GaInP 2 alloys

Leonid Chernyak, Andrei Osinsky, Henryk Temkin, Alexander Mintairov, I. G. Malkina, B. N. Zvonkov, and Yu. N. Saf'anov

Citation: Applied Physics Letters **70**, 2425 (1997); doi: 10.1063/1.118864
View online: http://dx.doi.org/10.1063/1.118864
View Table of Contents: http://scitation.aip.org/content/aip/journal/apl/70/18?ver=pdfcov
Published by the AIP Publishing



# Transport anisotropy in spontaneously ordered $GaInP_2$ alloys

Leonid Chernyak, Andrei Osinsky,[a)] and Henryk Temkin[b)]
*Department of Electrical Engineering, Texas Tech University, Lubbock, Texas 79409*

Alexander Mintairov
*A. F. Ioffe Physical-Technical Institute, 194021 Saint Petersburg, Russia*

I. G. Malkina, B. N. Zvonkov, and Yu. N. Saf'anov
*Physical-Technical Institute of State University, 603000 Nizhnii Novgorod, Russia*

(Received 16 December 1996; accepted for publication 4 March 1997)

Large anisotropy in minority carrier diffusion length and specific conductivity is observed in epitaxial layers of $GaInP_2$ alloys with $CuPt_B$-type ordering. Both the diffusion length and specific conductivity are enhanced, by factor of 10, in the [110] direction, parallel to the line of intersection of the ordered (1–11) and (−111) planes with the (001) growth surface. The reduction in transport length in the [1–10] direction is attributed to carrier scattering at domain boundaries. No transport anisotropy is observed in disordered $GaInP_2$ epitaxial layers. © *1997 American Institute of Physics.* [S0003-6951(97)02218-3]

Alloys of $GaInP_2$ grown by metal organic vapor phase epitaxy (MOVPE) on (001) GaAs are a remarkable example of spontaneous sublattice ordering.[1] Growth at high temperatures of 650–700 °C results in a monolayer superlattice (MSL) formed by alternating Ga- and In-rich layers along the ⟨111⟩ directions. Of the four possible equivalent directions, ordering is observed only along the [−111] and [1–11] directions, known as the $CuPt_B$ variants. The observed variants are caused by surface reconstruction effects.[2,3] On a macroscopic scale, the layers of $GaInP_2$ are partially ordered, having an ordering parameter $\eta \sim 0.2-0.5$, where $\eta$ is defined by the composition in the alternating planes of $Ga_{x+\eta/2}In_{1-(x+\eta/2)}P$ and $Ga_{x-\eta/2}In_{1-(x-\eta/2)}P$. Both $CuPt_B$ variants have common (−110) mirror planes. This leads to an optical anisotropy in the (001) growth plane of epitaxial layers. A strong anisotropy in ⟨110⟩ directions of $GaInP_2$ has been observed in polarized photoluminescence,[4] electroluminescence,[5] photocurrent,[6] and Raman spectra.[7] The effect of ordering on carrier transport is less well known. Most notably a mobility difference of 15% was observed between the [110] and [−110] directions in two-dimensional electron gas in $Ga_{0.25}In_{0.75}As/InP$ quantum wells.[8] This effect was attributed to either anisotropic ordering, such as spontaneous ordering forming a CuPt structure, or formation of In-rich stripes in the [−110] direction resulting from strained-layer epitaxy.

In this letter, we report large anisotropy of electrical conductivity and minority carrier diffusion length in (001) $GaInP_2$ layers. Samples, grown by MOVPE and molecular beam epitaxy (MBE), were characterized by Hall and current–voltage ($I-V$) measurements, atomic force microscopy (AFM), and electron beam induced current (EBIC) technique.

The layers of $GaInP_2$ were grown by atmospheric pressure MOVPE at 700 °C on semi-insulating (001) GaAs substrates with a 3° miscut towards the [110] direction. This tilt of the surface breaks the symmetry of the growing layer and results in the enhancement of the (1–11)-type variant, while the (−111) variant gradually disappears.[9] The V/III flux ratio was maintained at ~50. The $Ga(CH_3)_3$, $PH_3$ and $In(CH_3)_3$ reagents were used as sources of Ga, P, and In, respectively. All the samples were 1 $\mu$m thick. The lattice mismatch, determined by x-ray diffraction measurements, ranged from $-5\times10^{-3}$ to $7\times10^{-3}$. The ordering parameter $\eta$ was estimated from the ratio of intensities of the −1/2 1/2 1/2 superstructure reflection to the −3 3 3 structure reflection and was found to be in the range of 0.1–0.4.[10] The [110] and [1–10] directions within the growth plane were determined using chemical etching in a 0.5% $Br_2$: methanol solution.

The samples were doped $n$- and $p$-type with Si or Zn, respectively. The Ohmic contacts for Hall and $I-V$ measurements were prepared using In/Sn alloy and annealing at 300 °C for 10 min in $N_2$ atmosphere, or by welding tin strips by capacitance discharge. In $n$-type samples, the electron concentration ranged from $10^{16}$ to $10^{18}$ cm$^{-3}$, with the corresponding average electron mobilities in the 2800–680 cm$^2$/V s range; the $p$-type samples were doped from $3\times10^{17}$ to $2\times10^{18}$ cm$^{-3}$ with the corresponding average hole mobilities of 50–20 cm$^2$/V s. The unintentionally doped samples were $n$-type with the background electron concentration of $\sim10^{16}$ cm$^{-3}$ and average mobility $\mu = 2900$ cm$^2$/V s. In Si-doped samples, ordering decreases slowly with increasing doping level.[10] In Zn-doped samples, for $p<10^{18}$ cm$^{-3}$, the ordering is only weakly affected by the doping level.[11] While the transport anisotropy was observed in a variety of samples with $\eta=0.2-0.4$, the most prominent effect was measured in an unintentionally doped sample (to be referred below as the ordered sample) with the lattice mismatch $\sim7\times10^{-3}$ and $\eta=0.3$. The ordered $GaInP_2$ sample was compared with a reference $n$-type $GaInP_2$ epitaxial layer (referred to as the disordered sample) with a similar background carrier concentration. It was grown at 550 °C on (001) GaAs substrate by MBE.

The diffusion length of minority holes was investigated by electron beam induced current (EBIC) technique using a planar Schottky configuration.[12] The Schottky diodes were

a)Present address: APA Optics, 2950 NE 84th Lane, Minneapolis, MN 55449.
b)Author to whom correspondence should be addressed. Electronic mail: htemkin@coe2.coe.ttu.edu

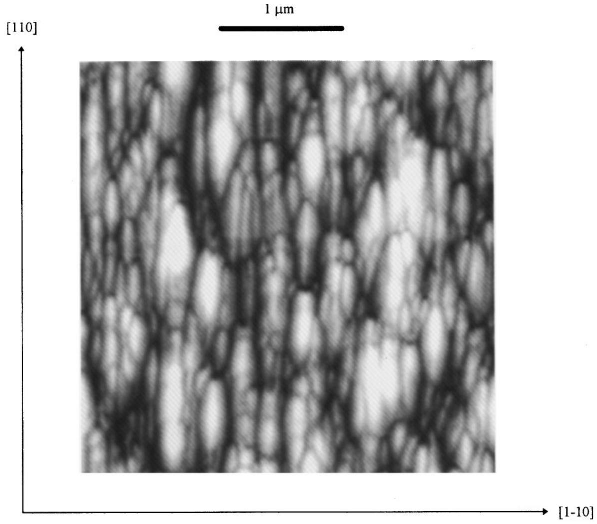


FIG. 1. The AFM image of surface morphology of ordered $GaInP_2$ epitaxial layer. The [110] and [1–10] directions are indicated by arrows.

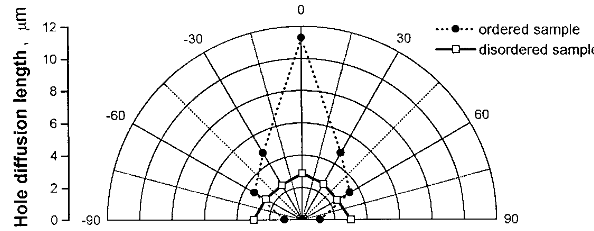


FIG. 2. Minority hole diffusion length dependence on the angle between the line created by the intersection of ordered (1–11) and (−111) planes with the (001) growth plane, and the electron beam scan direction. The angle was varied from −90° to 90° (the 0° direction corresponds to the scan direction parallel to the intersection line and coincides with the [110] direction).

formed by evaporation of 500 $\mu$m diameter and 150-nm-thick gold (Au) dots on the top surface of $GaInP_2$, aligned along [110] direction. As the electron beam scans away from the edge of the Au dot, the EBIC signal decays exponentially. A second Au dot served as the other contact. The minority hole diffusion length, $L$, was calculated by fitting the decay EBIC signal to the conventional expression for the current ($I$) dependence on the beam-to-dot distance ($d$), $I \sim \exp(-d/L)/d^{1/2}$.[12] Similar configuration was used for EBIC measurements in $CuInSe_2$ and Si.[13] Since the degree of ordering may change with depth within the epitaxial layer, low accelerating voltage (5 kV) of the scanning electron microscope (SEM) was used for EBIC measurements. Since the electron beam range (penetration depth) in the material ($\sim$0.3 $\mu$m)[12] is comparable to the width of the space charge region of the Schottky barrier ($\sim$0.4 $\mu$m), the use of low voltages may lead to an underestimate (by as much as $\sim$40%) of the minority carrier diffusion length.[12]

The influence of spontaneous ordering on the majority carrier (electron) transport was investigated by measuring the angular distribution of the dark $I-V$ curves taken between Ohmic contacts at the surface of $GaInP_2$ with respect to the line formed by the intersection of the (1–11) ordered planes and the (001) growth plane. Specific conductivity of the samples was calculated using $R=1/\sigma S$, where $R$ is the resistivity determined from Ohm's law; $\sigma$ is the specific conductivity; 1 is the distance between the centers of the contacts (precisely measured using the SEM); and $S$ is the cross section area of the electric current flux, essentially the same in every measurement.

The AFM plan-view image of the ordered sample (cf. Fig. 1) shows well-defined surface structure consisting of elongated regions aligned along the [110] direction. Similar AFM images of spontaneously ordered $GaInP_2$ have been reported recently by Nasi *et al.*[9] They observed a clear correspondence between the surface cathodoluminescence (CL) mapping of the ordered $GaInP_2$ samples and the topographic AFM images. The elongated bright and dark areas, imaged by CL, correspond to the ordered domains and the antiphase boundaries (APB), respectively.[9] The ondulations preferentially aligned along the [110] direction, revealed in AFM images, are ascribed to the absence of ordering in APBs. The reasons for the surface topography are not clear at present.

The aspect ratio of the elongated areas, seen in Fig. 1 (in the perpendicular [110] and [1–10] directions), ranges from 5:1 to 8:1. Our AFM investigation of the disordered sample does not show any specific morphological features. This is an additional argument in favor of a correlation between the surface morphology and ordering. Since the APB ridges seen in Fig. 1 are, respectively, parallel and perpendicular to the [110] and [1–10] directions, we expect different electrical transport properties in these directions. Indeed, such a transport anisotropy is observed both for the minority and majority carriers, expressed in the minority carrier (hole) diffusion length and specific conductivity, respectively.

The angular dependence of the hole diffusion length in ordered and disordered samples is presented in Fig. 2. In the ordered sample, the angle between the line, created by the intersection of ordered (1–11) and (−111) planes with the (001) growth surface, and the electron beam scan direction was varied from −90° to 90° (the 0° direction corresponds to the electron beam that is parallel to the intersection line and coincides with the [110] direction; the 90° direction is perpendicular to the intersection line and coincides with the [1–10] direction). One can see a very strong anisotropy in the diffusion length in the ordered sample, exceeding 10:1, for the [110] and [1–10] directions, respectively. The maximum $L$ is so large that it becomes comparable to the electron diffusion length in conventional III–V compounds. For instance, a similar $L$ was measured in $p$-type GaInAs having a hole concentration in the range of $10^{16}-10^{17}$ cm$^{-3}$.[14] In the case of the disordered sample, the hole diffusion length is isotropic and much smaller (cf. Fig. 2).

Figure 3 presents the angular variation of $\sigma S$ for the ordered and disordered samples. The angle was again measured between the line created by intersection of the ordered (1–11) and (−111) planes with the (001) growth surface, and the direction of electric current flux (where the 0° and 90° directions are as defined above). One can see (cf. Fig. 3) a strong anisotropy of specific conductivity in the ordered sample. In terms of the angular dependence, a maximum of $\sigma S$ coincides with that of the minority hole diffusion length and is parallel to the [110] direction. This is apparently a direction in which both the majority and minority carrier scattering is minimal. No conductivity anisotropy was ob-

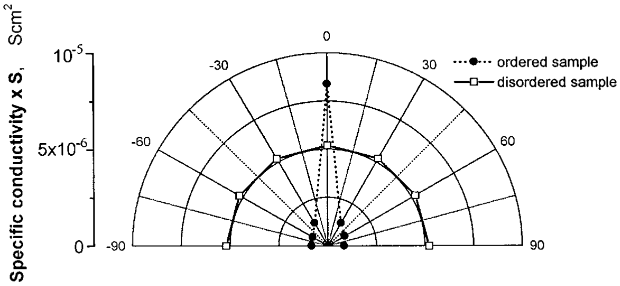


FIG. 3. Angular variation of the product $\sigma S$, where the angle is measured between the line created by the intersection of ordered (1–11) and (−111) planes with the (001) growth plane, and the direction of electric current flow. The angles of 0° and 90° correspond to the [110] and [1–10] directions, respectively.

served for the disordered sample. It might be interesting to point out that carrier scattering at antiphase boundaries has been invoked previously in studies of ordering and the associated electrical resistivity changes in $Cu_3Au$.[15] The increase in the electrical conductivity with the domain size is attributed to conduction electron scattering at antiphase domain boundaries.[15]

We attribute the observed transport anisotropy to the ordered domain boundary scattering. As is evident from the CL mapping[9] and, supported by the AFM image of the ordered sample (cf. Fig. 1), the domain boundary density is much higher in the [1–10] than in the [110] direction. The carrier scattering in the [110] direction is therefore expected to be less pronounced. We note that our results are consistent with previously published data on the carrier mobility decrease in the ordered $GaInP_2$ epilayers.[16] The decrease of carrier mobility can be related to recombination and scattering at domain boundaries as well as, to some degree, to phase separation.[17] We note, in addition, that the higher band gap associated with domain boundaries could act as a barrier and, therefore, affect the diffusion length.[18,19]

The research at Texas Tech was supported by the J.S Maddox Foundation. The N. Novgorod group was supported by the RFBR 95-02-05610, Physics of Solid State Nanostructures Program 2-193, and INTAS 93-1403.

[1] A. Zunger and S. Mahajan, Hand. Semicond. **3**, 1399 (1994).
[2] A. Gomyo, K. Maakita, I. Hino, and T. Suzuki, Phys. Rev. Lett. **72**, 673 (1994).
[3] S. B. Zang, S. Froyen, and A. Zunger, Appl. Phys. Lett. **67**, 3141 (1995).
[4] A. C. Chen, A. M. Moy, L. J. Chou, K. C. Hsieh, and K. Y. Cheng, Appl. Phys. Lett. **66**, 2694 (1995).
[5] E. Greger, K. H. Gulden, P. Riel, H. P. Schweizer, M. Moser, G. Schmiedel, P. Kiesel, and G. H. Döhler, Appl. Phys. Lett. **68**, 2383 (1996).
[6] T. Kita, A. Fujiwara, H. Nakayama, and T. Nishino, J. Electron. Mater. **25**, 661 (1996).
[7] A. M. Mintairov and V. G. Melehin, Semicond. Sci. Technol. **11**, 904 (1996).
[8] P. Ramvall, N. Carlsson, P. Omling, L. Samuelson, W. Seifert, M. Stolze, and Q. Wang, Appl. Phys. Lett. **68**, 1111 (1996).
[9] L. Nasi, G. Salviati, M. Mazzer, and C. Zanotti-Fregonara, Appl. Phys. Lett. **68**, 3263 (1996).
[10] Yu. N. Saf'anov, Crystallogr. Rep. **41**, 335 (1996), in Russian.
[11] A. Gomyo, H. Hotta, I. Hino, S. Kawata, K. Kobayashi, and T. Suzuki, Jpn. J. Appl. Phys. **28**, L1330 (1989).
[12] D. E. Ioannou and C. A. Dimitriadis, IEEE Trans. Electron Devices **29**, 445 (1996); C. A. Dimitriadis, J. Phys. D **14**, 2269 (1981); H. J. Leamy, J. Appl. Phys. **53**, R51 (1982).
[13] L. Chernyak, K. Gartsman, D. Cahen, and O. M. Stafsudd, J. Phys. Chem. Solids **56**, 1165 (1995); L. Chernyak, V. Lyakhovitskaya, Sh. Richter, A. Jakubowicz, Y. Manassen, S. R. Cohen, and D. Cahen, J. Appl. Phys. **80**, 2749 (1996).
[14] M. B. Panish and H. Temkin, *Gas Source Molecular Beam Epitaxy* (Springer, Berlin 1993).
[15] P. L. Rossiter and B. Bykoves, Philos. Mag. B **38**, 555 (1978); F. W. Jones and C. Sykes, Proc. R. Soc. London, Ser. A. **157**, 213 (1936).
[16] D. J. Friedman, A. E. Kibbler, and J. M. Olson, Appl. Phys. Lett. **59**, 2998 (1991).
[17] M. A. Shahid and S. Mahajan, Phys. Rev. B **38**, 1344 (1988).
[18] M. C. DeLong, W. D. Ohlsen, I. Viohi, P. C. Taylor, and J. M. Olson, J. Appl. Phys. **70**, 2780 (1991).
[19] J. E. Fouquet, M. S. Minsky, and S. J. Rosner, Appl. Phys. Lett. **63**, 3212 (1993).